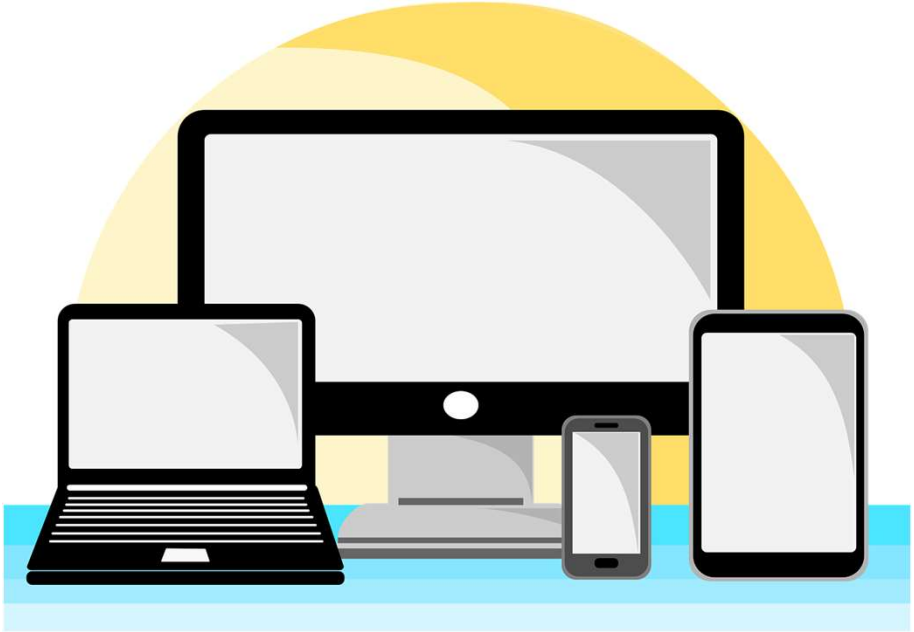
System and Devices (SYS 3)

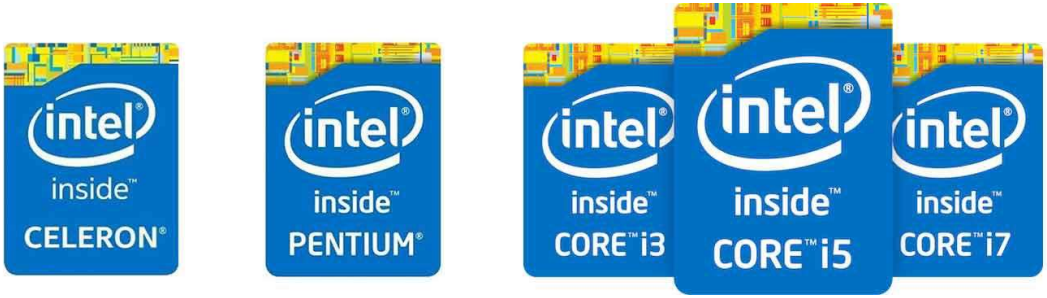
Lecture 1: Introduction to Advanced Computer Architecture

Dr Pengcheng Liu

Computer Architects: Its Role in Real-Life

Hand-held devices and their applications are part of our day-to-day life.



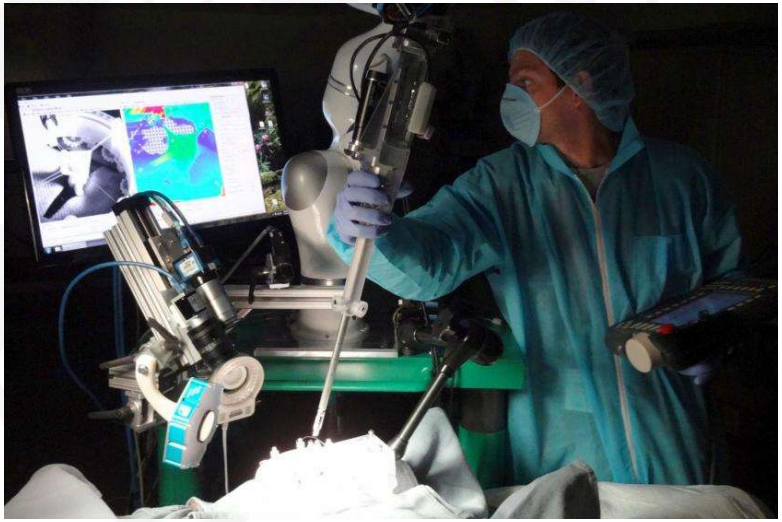


What are the key applications in future decades that need high end architectural support to sustain?

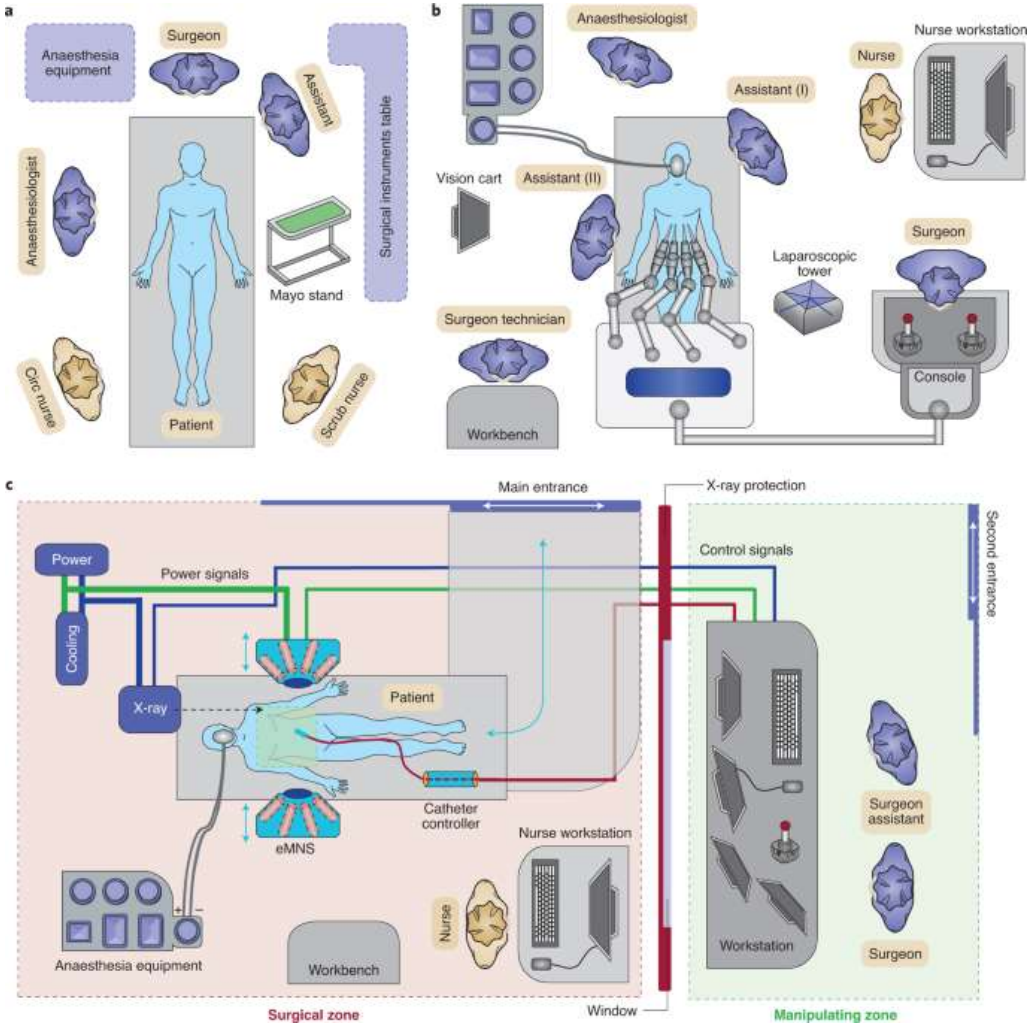
Precision Agriculture

Smart Surgery and Healthcare

Source: https://www.sjchs.org/living-smart-blog/blog-details/blog/2019/01/10/what-is-robot-assisted-surgery

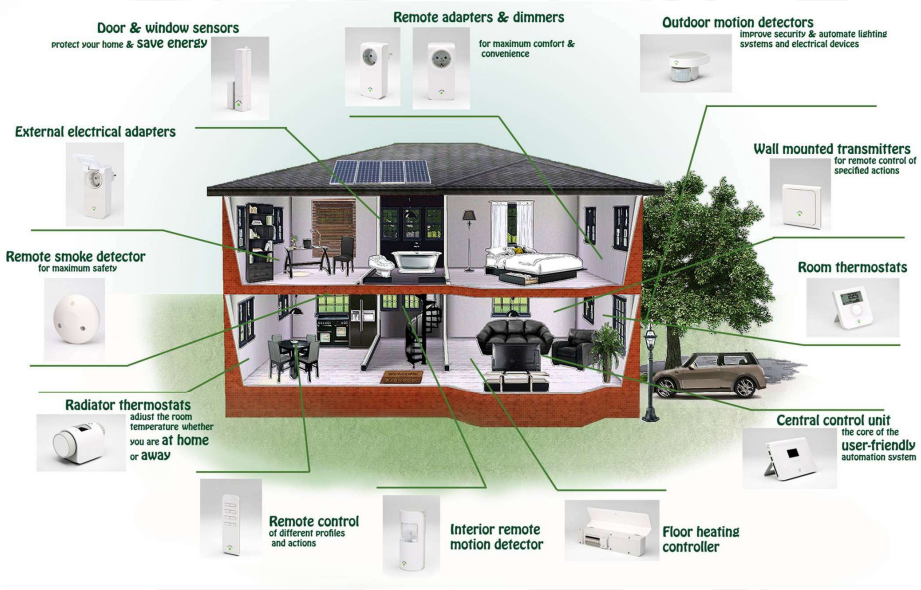


Source: https://www.medicaldaily.com/robot-surgeon-smart-tissue-autonomous-386079



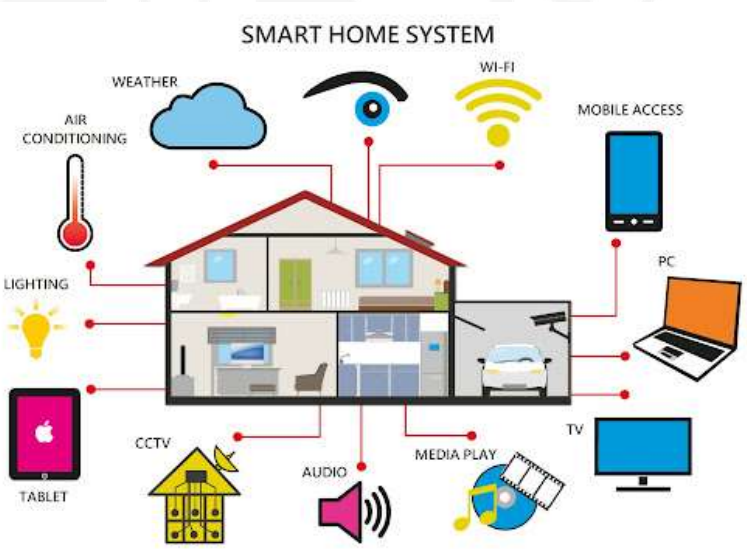
Zemmar, A., Lozano, A.M. and Nelson, B.J., 2020. The rise of robots in surgical environments during COVID-19. Nature Machine Intelligence, 2(10), pp.566-572.

Smart Homes



Source: https://www.refitsmarthomes.org/

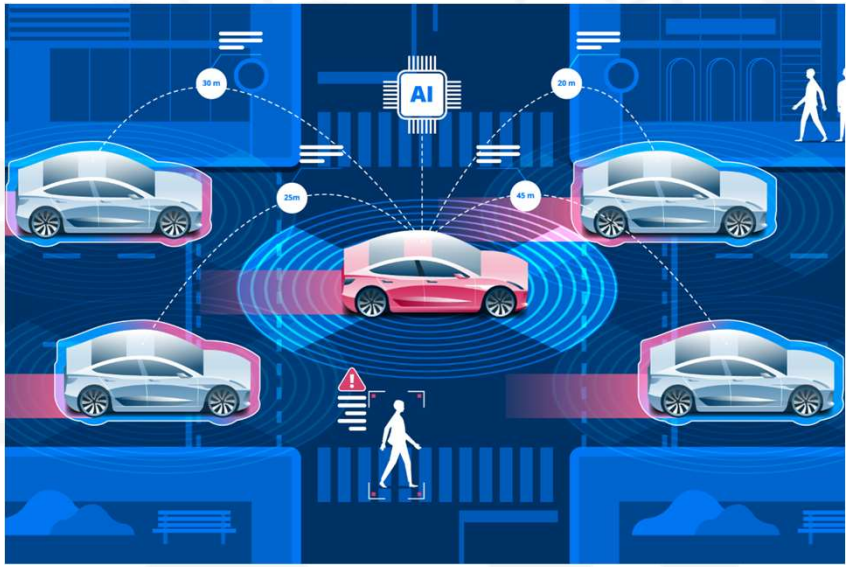
Source: https://ubuntumanual.org/smart-homes-gadgets/



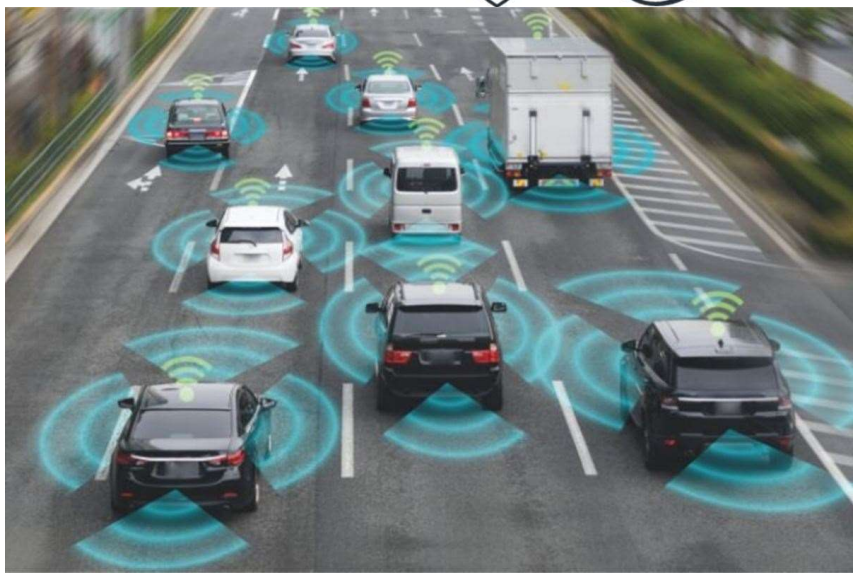
Source: http://visioforce.com/smarthome.html

Source: https://www.cbinsights.com/research/drone-impact-society-uav/

Smart Transportation Systems

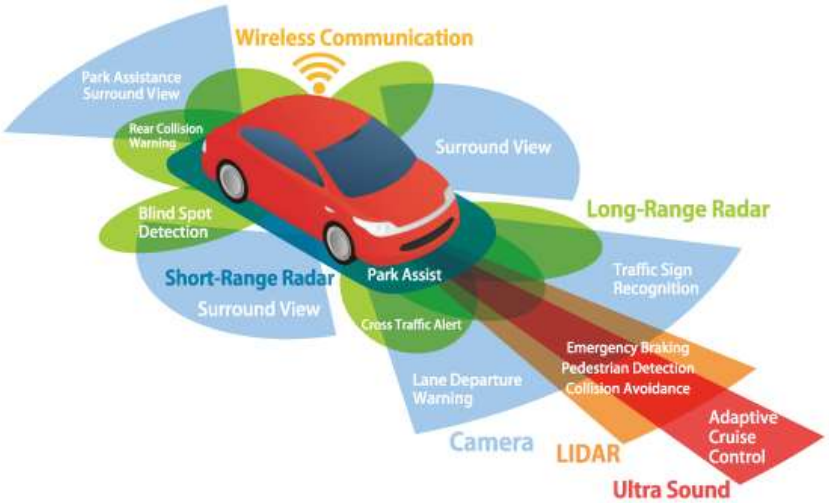


Source: https://www.smartcitiesworld.net/opinions/opinions/driving-autonomous-vehicles-forward-with-intelligent-infrastructure



Source: https://www.theparliamentmagazine.eu/news/article/autonomous-driving-a-glimpse-into-the-future

Source: https://www.kaspersky.co.uk/blog/secure-futures-magazine/cybersecurity-autonomous-vehicles/20332/



Source: https://ldcinfrastructure.com.au/autonomous-car-3/

TEF Gold Teaching Excellence Framework

High Precision Security Devices

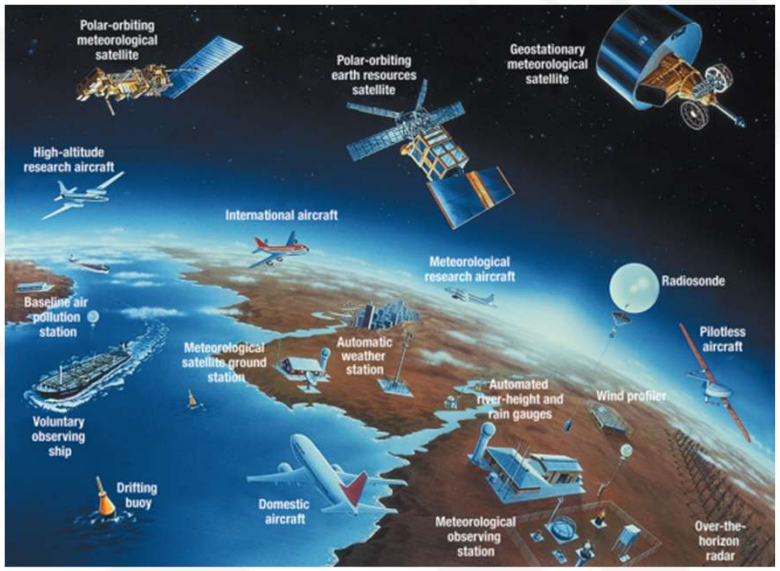
Source: https://pixabay.com/photos/jobsite-security-camera-2427499/

Source: https://threatpost.com/drones-breach-cyberdefenses/143075/

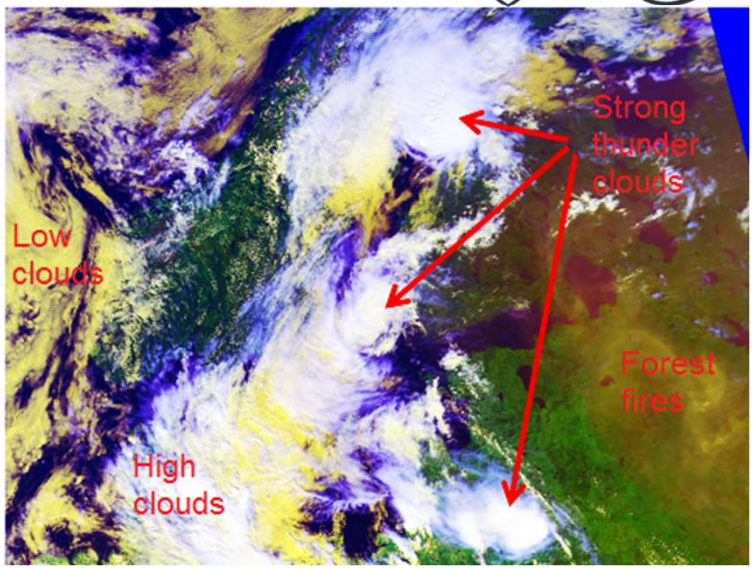
Source: https://unsplash.com/s/photos/facial-recognition

Source: https://www.security.org/resources/smart-home-automation-guide/

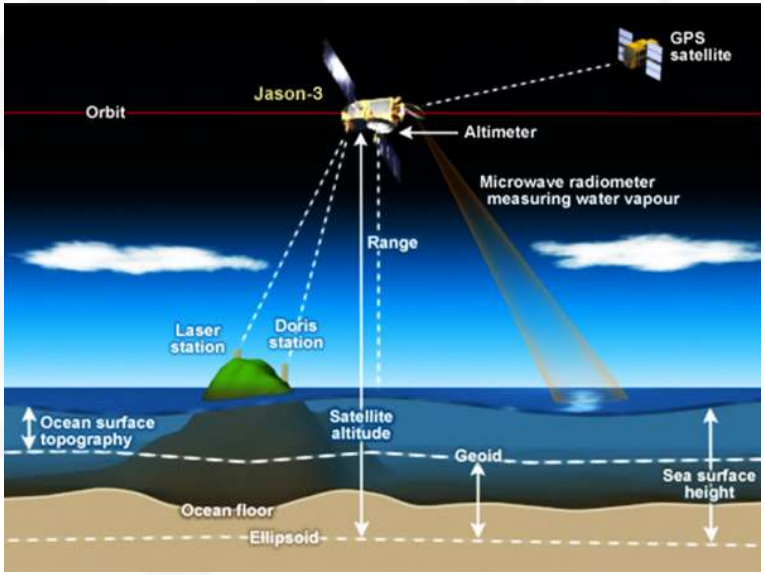
Weather Forecasting



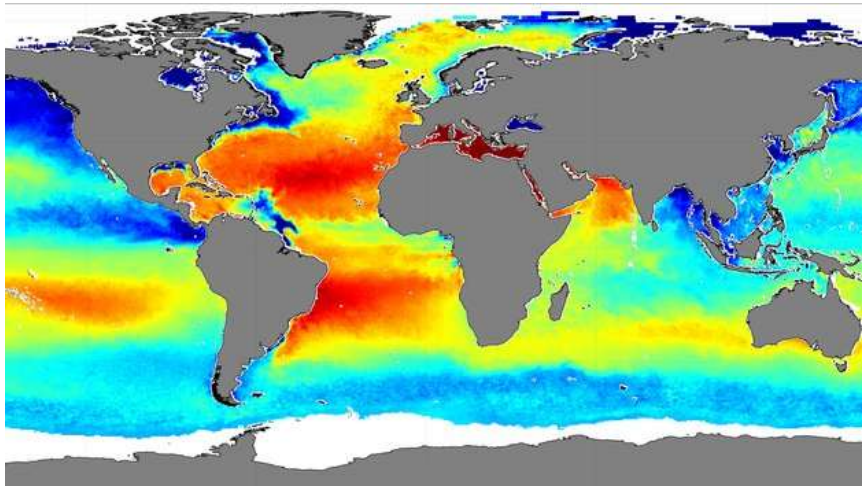
Source: https://public.wmo.int/en/about-us/vision-and-mission/wmo-integrated-global-observing-system



Source: https://en.ilmatieteenlaitos.fi/satellite-observations-of-weather



Source: https://milesobrien.com/satellites-track-weather-climate-change/



Source: https://www.geospatialworld.net/blogs/satellites-for-monitoring-climate-change/

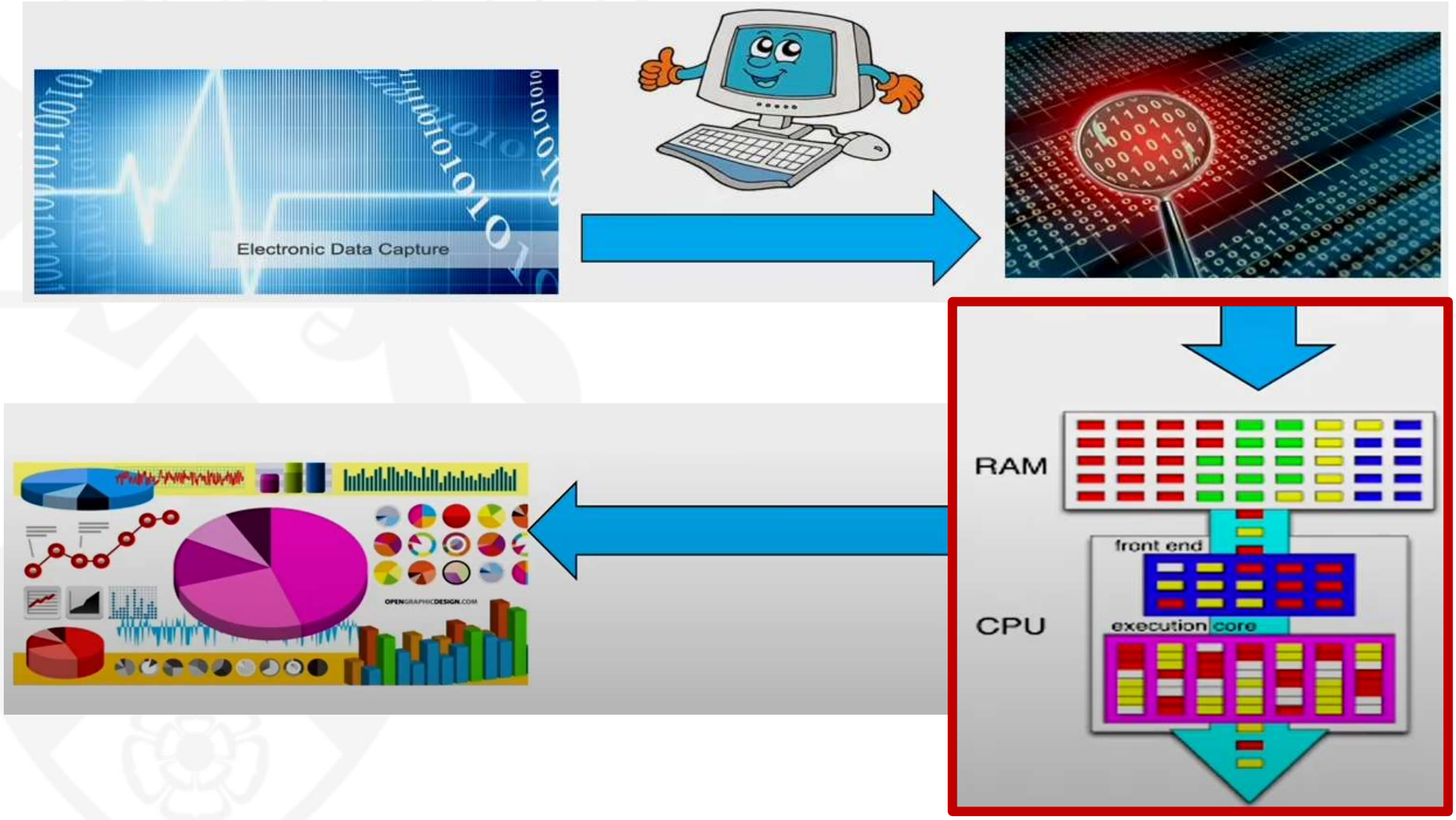
Smart City to Smart Planet

Source: https://internetofthingsagenda.techtarget.com/definition/smart-city

Source: https://www.brandingmag.com/2011/05/15/ibm-a-smarter-planet/

SmartPlanet

How is this all done?



Summary